

Setting Priors in brms

Bayesian Mixed Effects Models with brms for Linguists

Workshop Materials

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1 Setting Priors in brms (20 min)

1.1 Default vs. Weakly Informative Priors

brms uses weakly informative priors by default (not completely flat). However, for psycholinguistics, domain-specific priors are even better.

1.1.1 What is a Prior?

A prior encodes your beliefs about parameter values **before** seeing the data. In Bayesian inference:

$$\text{posterior} \propto \text{likelihood} \times \text{prior}$$

Types of priors:

- **Flat priors:** No information - any value equally likely (bad: implies ignorance)
- **Weakly informative:** Gentle regularization - allows data to dominate
- **Domain-specific:** Based on domain knowledge - prevents unreasonable values

1.1.2 The Intercept Adapts to Your Data!

This is important: The default intercept prior depends on `mean(y)`. Let's see this in action:

```
library(brms)
library(tidyverse)

# Create example datasets with different scales
set.seed(123)

# RT data: log-transformed, mean 6 ( 400ms)
rt_data_typical <- data.frame(
  subject = factor(rep(1:20, each = 10)),
  item = factor(rep(1:10, times = 20)),
  condition = factor(rep(c("A", "B"), each = 5, times = 20)),
  log_rt = rnorm(200, mean = 6, sd = 0.5)
)

# RT data: extreme scale, mean 10 ( 22,000ms - unrealistic)
rt_data_extreme <- data.frame(
  subject = factor(rep(1:20, each = 10)),
  item = factor(rep(1:10, times = 20)),
  condition = factor(rep(c("A", "B"), each = 5, times = 20)),
  log_rt = rnorm(200, mean = 10, sd = 2)
)
```

Now let's check what default priors brms suggests:

```
cat("\n=== DEFAULT PRIORS: Typical RT data (mean log-RT 6) ===\n\n")
```

```
=== DEFAULT PRIORS: Typical RT data (mean log-RT 6) ===
```

```
rt_priors_typical <- get_prior(
  log_rt ~ condition + (1 + condition | subject) + (1 | item),
  data = rt_data_typical,
  family = gaussian())
```

```
)
print(rt_priors_typical)
```

	prior	class	coef	group	resp	dpar	nlpar	lb	ub	tag
	(flat)	b								
	(flat)	b	conditionB							
	lkj(1)	cor								
	lkj(1)	cor		subject						
student_t(3, 6, 2.5)		Intercept								
student_t(3, 0, 2.5)		sd							0	
student_t(3, 0, 2.5)		sd			item				0	
student_t(3, 0, 2.5)		sd	Intercept		item				0	
student_t(3, 0, 2.5)		sd		subject					0	
student_t(3, 0, 2.5)		sd	conditionB	subject					0	
student_t(3, 0, 2.5)		sd	Intercept	subject					0	
student_t(3, 0, 2.5)		sigma							0	
source										
default										
(vectorized)										
default										
(vectorized)										
default										
default										
(vectorized)										
(vectorized)										
(vectorized)										
(vectorized)										
(vectorized)										
default										

```
cat("\n\n=== DEFAULT PRIORS: Extreme RT data (mean log-RT 10) ===\n\n")
```

```
=== DEFAULT PRIORS: Extreme RT data (mean log-RT 10) ===
```

```
rt_priors_extreme <- get_prior(
  log_rt ~ condition + (1 + condition | subject) + (1 | item),
  data = rt_data_extreme,
  family = gaussian())
```

```
)
print(rt_priors_extreme)
```

```

      prior      class      coef  group resp dpar nlpar lb ub tag
      (flat)         b
      (flat)         b conditionB
      lkj(1)        cor
      lkj(1)        cor          subject
student_t(3, 10, 2.5) Intercept
student_t(3, 0, 2.5)      sd                      0
student_t(3, 0, 2.5)      sd          item          0
student_t(3, 0, 2.5)      sd Intercept  item          0
student_t(3, 0, 2.5)      sd          subject          0
student_t(3, 0, 2.5)      sd conditionB subject          0
student_t(3, 0, 2.5)      sd Intercept subject          0
student_t(3, 0, 2.5)      sigma                    0
      source
      default
(vectorized)
      default
(vectorized)
      default
      default
(vectorized)
(vectorized)
(vectorized)
(vectorized)
(vectorized)
      default
```

```
# Compare intercept priors
cat("\n=== Intercept Comparison ===\n")
```

```
=== Intercept Comparison ===
```

```
cat("Typical data prior: ",
    rt_priors_typical[rt_priors_typical$class == "Intercept", "prior"], "\n")
```

```
Typical data prior:  student_t(3, 6, 2.5)
```

```
cat("Extreme data prior: ",  
    rt_priors_extreme[rt_priors_extreme$class == "Intercept", "prior"], "\n")
```

Extreme data prior: student_t(3, 10, 2.5)

```
cat("→ The intercept prior CHANGES with data scale!\n")
```

→ The intercept prior CHANGES with data scale!

Key insight: The intercept prior automatically scales with your data. This is convenient but has a problem: **if you don't specify priors, your prior assumptions implicitly depend on how you code your variables!**

1.2 Default brms Priors

When you don't specify priors, brms assigns defaults:

- **Intercept:** `student_t(3, mean(y), 2.5)` - **DATA-DEPENDENT!** Centers at your data mean
 - Adapts to your data scale automatically
 - For RT data with `mean(log_rt) = 6`: allows roughly 150ms-1100ms range
- **Slopes (b):** `(flat)` - improper uniform prior over $(-\infty, +\infty)$
 - No information: any effect size equally likely
 - Technically improper (doesn't integrate to 1)
- **Sigma** (residual SD): `student_t(3, 0, 2.5)` with lower bound 0
 - Weakly informative for residual variance
- **SD** (random effects): `student_t(3, 0, 2.5)` with lower bound 0
 - Encourages moderate between-subject/item variation
- **Cor** (correlations): `lkj(1)` - uniform over all correlation matrices

1.3 Setting Weakly Informative Priors for Reaction Times

For psycholinguistics, it's better to specify priors based on domain knowledge:

```
# Define priors explicitly
rt_priors <- c(
  prior(normal(6, 1.5), class = Intercept, lb = 4), # log(RT) around 400ms, min ~55ms
  prior(normal(0, 0.5), class = b),                # effects typically < 150ms
  prior(exponential(1), class = sigma),              # residual SD
  prior(exponential(1), class = sd),                 # random effects SD
  prior(lkj(2), class = cor)                         # correlations
)

cat("Our weakly informative priors:\n")
```

Our weakly informative priors:

```
print(rt_priors)
```

	prior	class	coef	group	resp	dpar	nlpar	lb	ub	tag	source
	normal(6, 1.5)	Intercept						4	<NA>		user
	normal(0, 0.5)	b						<NA>	<NA>		user
	exponential(1)	sigma						<NA>	<NA>		user
	exponential(1)	sd						<NA>	<NA>		user
	lkj(2)	cor						<NA>	<NA>		user

1.3.1 Why These Numbers?

1.3.1.1 normal(6, 1.5) for Intercept with lb = 4

- **Mean = 6** on log scale $\rightarrow \exp(6)$ 403ms (typical RT)
- **SD = 1.5** $\rightarrow$ 95% prior interval spans [3, 9] on log scale
- **Lower bound = 4** $\rightarrow \exp(4)$ 55ms (minimum physiologically possible motor response)
 - Without lb, the prior puts only ~9% probability below this threshold
 - Adding lb makes our assumption explicit: we don't entertain impossibly fast RTs
 - In practice, data will dominate anyway, but bounded priors clarify our theoretical assumptions
- But 95% of prior **mass** is between $\pm 1.96 \times 1.5$ around the mean
- This puts most probability on reasonable RTs (100-1500ms), while still allowing outliers

1.3.1.2 `normal(0, 0.5)` for Effects

- **Mean = 0** (no directional assumption)
- **SD = 0.5** on log scale
- 95% prior interval: $[-1, 1]$ on log scale
- Translates to effect sizes of roughly $\pm 65\%$ of baseline (multiplicative)
- Or approximately $\pm 100\text{-}150\text{ms}$ for typical RTs around $400\text{-}600\text{ms}$
- **Why not flat?** Flat priors prefer extreme effect sizes (counterintuitive!)

1.3.1.3 `exponential(1)` for Sigma and SD

- Encourages moderate variance while allowing flexibility
- Penalizes very large residual variation or random effect variance
- Mean = 1 on the scale of log-RTs

1.3.1.4 `lkj(2)` for Correlations

- $= 2$: slight preference for correlations near 0
- Skeptical of strong correlations (like perfect intercept-slope correlation)
- If you truly expected strong correlations, you could use `lkj(1)` or lower

1.3.2 Comparison: Normal vs. Student-t

brms defaults use `student_t(3, , 2.5)` which has heavier tails than `normal()`. Why switch?

```
# Compare tail behavior
set.seed(456)
n_samples <- 100000
normal_samples <- rnorm(n_samples, 0, 0.5)
studentt_samples <- rt(n_samples, 3) * 0.5

normal_99 <- quantile(normal_samples, c(0.001, 0.999))
studentt_99 <- quantile(studentt_samples, c(0.001, 0.999))

cat("Tails comparison (1st and 99th percentiles):\n")
```

Tails comparison (1st and 99th percentiles):

```
cat("Normal(0, 0.5):      ", round(normal_99, 2), "\n")
```

Normal(0, 0.5): -1.55 1.56

```
cat("Student-t(3) × 0.5:   ", round(studentt_99, 2), "\n")
```

Student-t(3) × 0.5: -5.26 5.16

```
cat("Student-t ratio:      ", round(studentt_99[2] / normal_99[2], 2), "x wider\n\n")
```

Student-t ratio: 3.3 x wider

```
cat("Student-t allows extreme values 2x more likely than normal!\n\n")
```

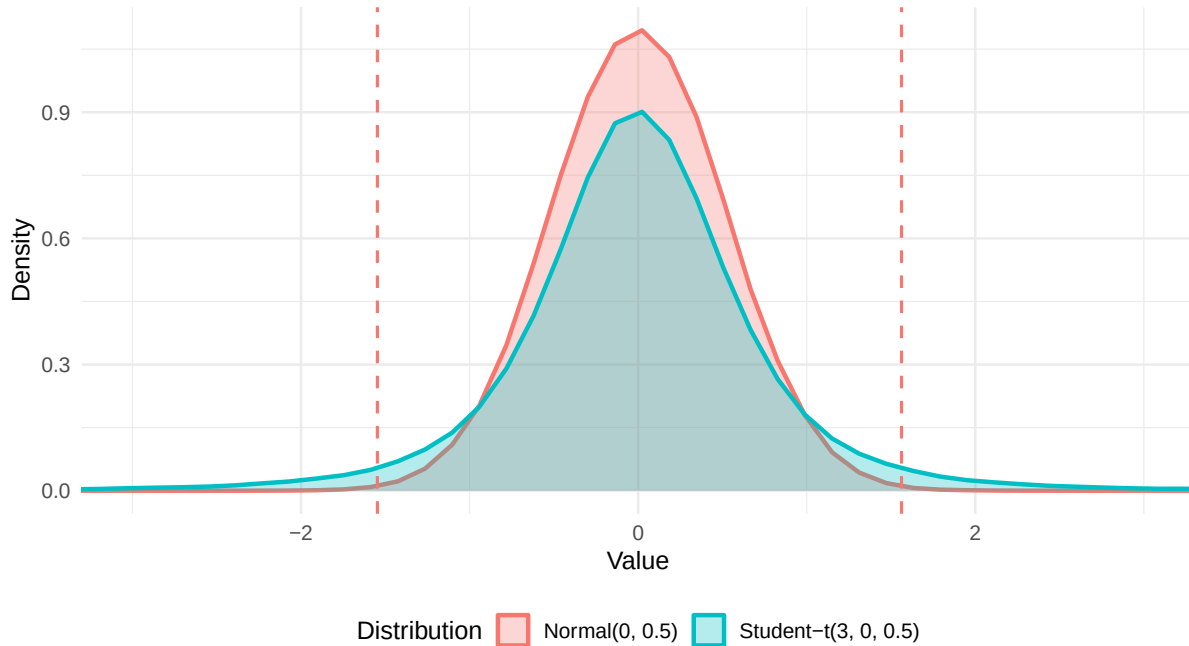
Student-t allows extreme values 2x more likely than normal!

```
# Graphical comparison
library(ggplot2)
comparison_df <- data.frame(
  value = c(normal_samples, studentt_samples),
  distribution = rep(c("Normal(0, 0.5)", "Student-t(3, 0, 0.5)"), each = n_samples)
)

ggplot(comparison_df, aes(x = value, fill = distribution, color = distribution)) +
  geom_density(alpha = 0.3, linewidth = 1) +
  geom_vline(xintercept = normal_99, linetype = "dashed", color = "#F8766D", linewidth = 0.7) +
  geom_vline(xintercept = studentt_99, linetype = "dashed", color = "#00BFC4", linewidth = 0.7) +
  coord_cartesian(xlim = c(-3, 3)) +
  scale_fill_manual(values = c("#F8766D", "#00BFC4")) +
  scale_color_manual(values = c("#F8766D", "#00BFC4")) +
  labs(
    title = "Normal vs. Student-t Prior Distributions",
    subtitle = "Student-t has heavier tails (dashed lines = 0.1% and 99.9% quantiles)",
    x = "Value",
    y = "Density",
    fill = "Distribution",
    color = "Distribution"
  ) +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom")
```


Normal vs. Student-t Prior Distributions

Student-t has heavier tails (dashed lines = 0.1% and 99.9% quantiles)



When to use each: - **Student-t:** Default choice (conservative, robust to outliers) - **Normal:** When you have strong domain knowledge about plausible ranges (better for RT data in controlled experiments)

Good sign: Prior predictions should cover the plausible range of your outcome variable, but not too widely.

1.4 Summary

Key takeaways:

1. **Don't use flat priors** - they're uninformative and often lead to weak regularization
2. **Default intercept priors adapt to data** - implicit assumptions depend on your coding!
3. **Specify priors explicitly** based on domain knowledge
4. **Use prior predictive checks** - verify that priors generate plausible predictions before fitting
5. **Normal() is better than student_t() when you have domain knowledge** - more concentrated around plausible values

Next steps: - See 02_prior_predictive_checks_rt.qmd for detailed prior validation - See 03_posterior_predictive_checks_rt.qmd for checking if the fitted model makes sense